

JMORF/MORS — Morpho-Syntax

Non-referential NPs, Expletives, and Extraposition

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Lecture 10

Location: KB-1.11

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Overview

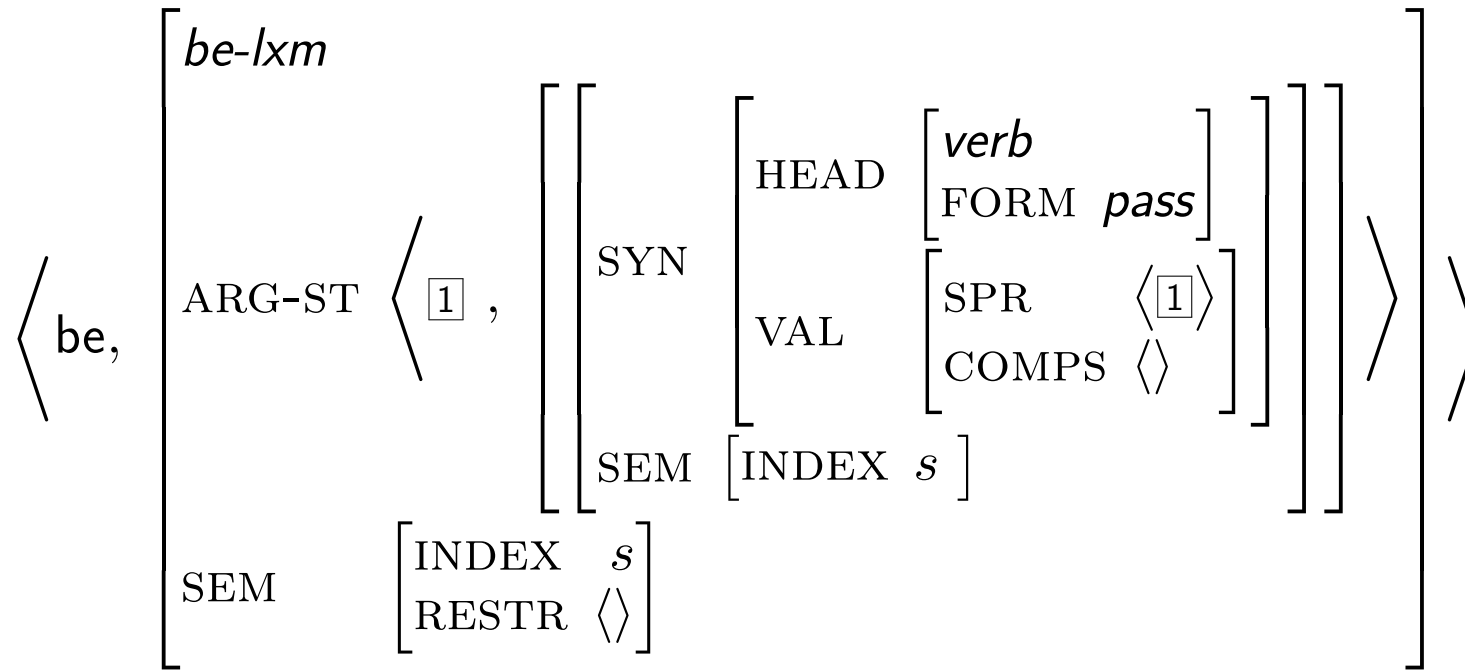
- Review: the copula *be*
- Existentials (*There is an X, ...*)
- Dummy *it*
- Complementizers and CP subjects
- Extraposition (*It worries me that X, ...*)
- Idioms (*X takes advantage of Y, ...*)
- Problems

Review

Introduction

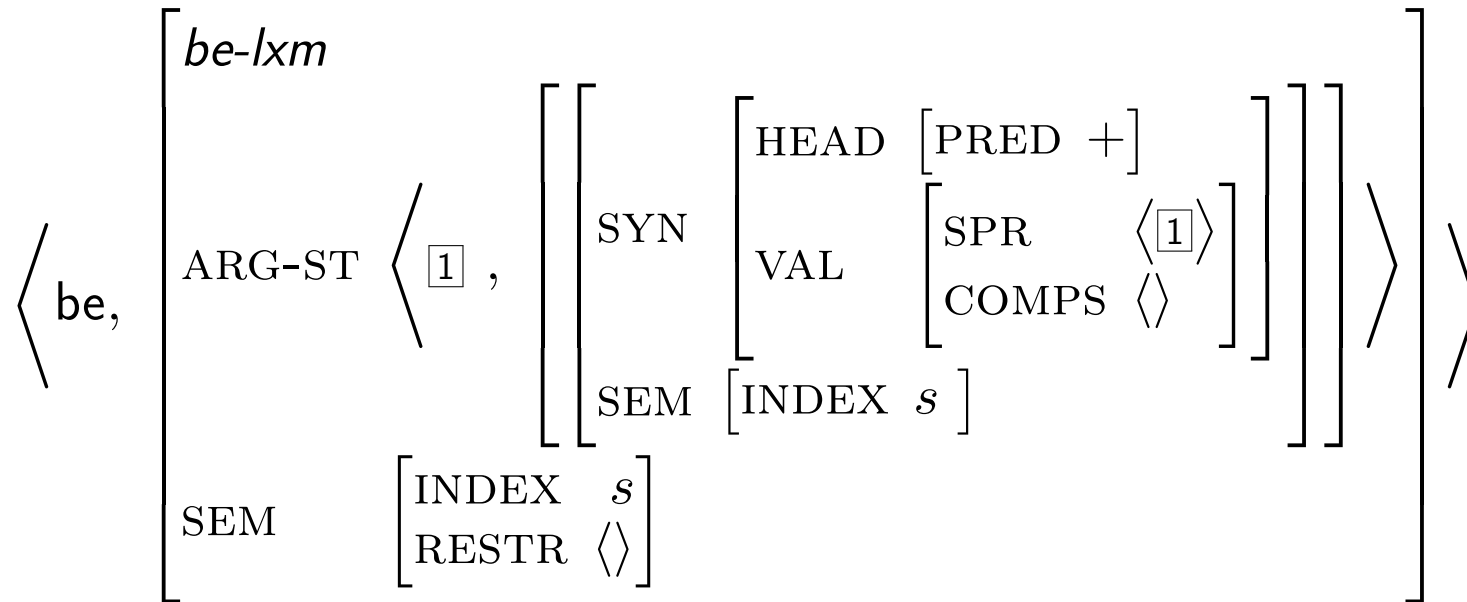
- In Chapter 10, we met the passive *be*.
- Passive *be* is just a special case — that *be* generally introduces [PRED +] constituents.
- Today, we'll introduce another *be*, which occurs in existential sentences starting with *there*, e.g. *There is a monster in Loch Ness*.
- Then we'll look at this use of *there*.
- Which will lead us to a more general examination of NPs that don't refer, including some uses of *it* and certain idiomatic uses of NPs.

Chapter 10 Entry for *be*



This takes only passive lexemes: $[\text{FORM } \textit{pass}]$.

Copula (Generalized)



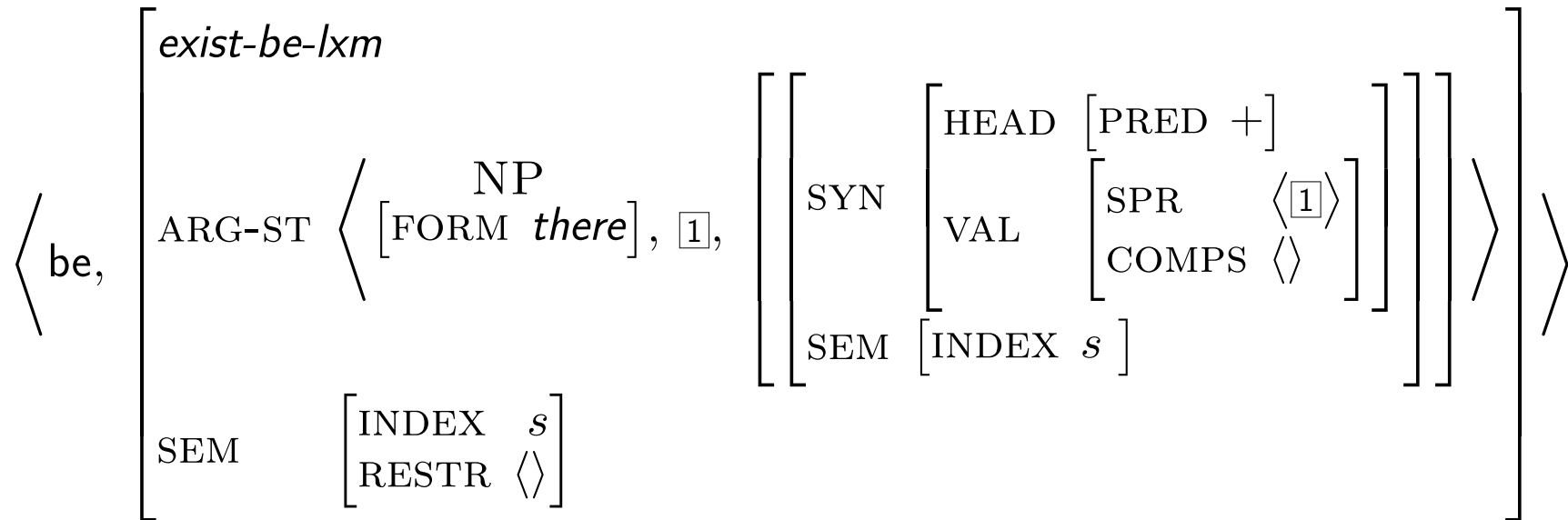
- Allow also (**predicative**) adjectives *fond*, *sleeping* are [PRED +]
- *mere* is [PRED −]
- Most adjectives can be either

Existentials

Existentials

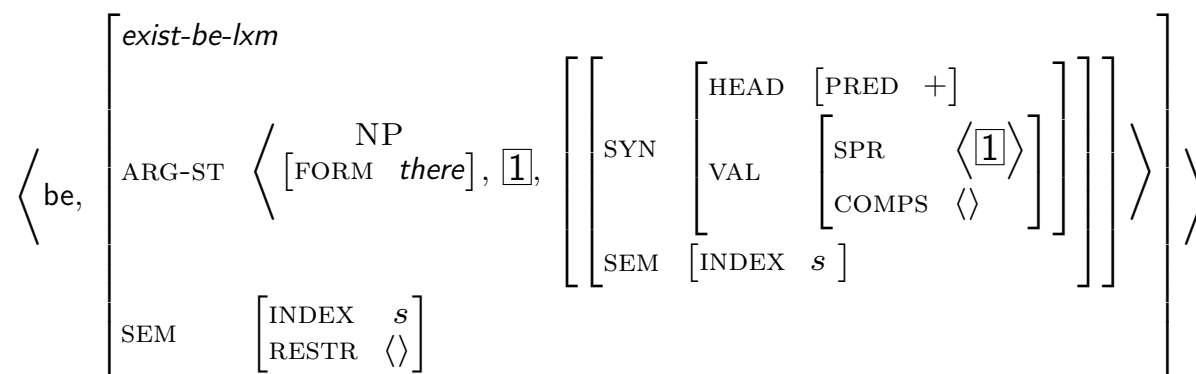
- The *be* in *There is a page missing* cannot be the same *be* that occurs in sentences like *Pat is tall* or *A cat was chased by a dog*. Why not?
- So we need a separate lexical entry for this *be*, stipulating:
 - Its SPR must be *there*.
 - It takes two complements, the first an NP and the second an AP, PP, or (certain kind of) VP.
 - The semantics should capture the relation between, e.g. *There is a page missing* and *A page is missing*.
- More examples:
 - (1) *There was a person at the door.*
 - (2) *There are many people fond of linguistics.*
 - (3) *There are people looking at us.*

Lexical Entry for the Existential *be*



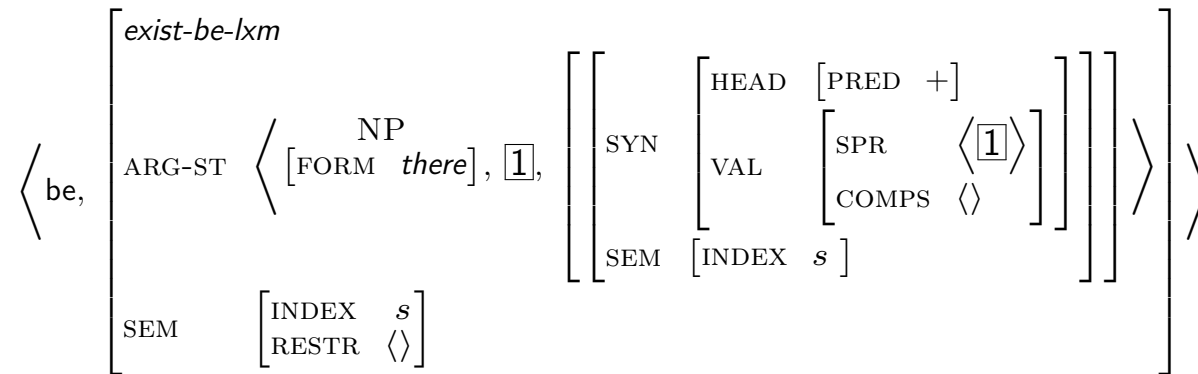
Questions About the Existential *be*

- What type of constituent is the third argument?
- Why is the third argument [PRED +]?
- Why is the second argument tagged as identical to the SPR of the third argument?



There Are Questions Left

- What is the contribution of this *be* to the semantics of the sentences it occurs in?
- Can all [PRED +] predicates appear as the third argument in existentials?
- How do we rule out **There was a greyhound a good runner*?



The Entry for Existential *there*

$\langle$ there,	<i>pron-lxm</i>		
	SYN	HEAD	$\left[\begin{array}{l} \text{FORM } \textit{there} \\ \text{AGR } \left[\text{PER } \textit{3rd} \right] \end{array} \right]$
	SEM	MODE	<i>none</i>
		INDEX	<i>none</i>
		RESTR	$\langle \rangle$

Questions About Existential *there*

- Why do we call it a pronoun?
- Why don't we give it a value for NUM?
- What does this entry claim is *there*'s contribution to the semantics of the sentences it appears in?
- Is this a correct claim?

$\langle$ there,	<i>pron-lxm</i>		
	SYN	HEAD	$\left[\begin{array}{l} \text{FORM } there \\ \text{AGR } [PER \ 3rd] \end{array} \right]$
	SEM	$\left[\begin{array}{l} \text{MODE } none \\ \text{INDEX } none \\ \text{RESTR } \langle \rangle \end{array} \right]$	$\rangle$

Dummy *it*

Other NPs That Don't Seem to Refer

- (4) *It sucks that the Rockies lost the series.*
- (5) *It is raining.*
- (6) *Andy took advantage of the opportunity.*
- (7) *Lou kicked the bucket.*

What About *It follows that you are wrong*?

- This is an example of **extraposition**.
- To analyze it, we need:
 - An analysis of this use of *that**
 - Entries for verbs that take clausal subjects*
 - (8) *That you are wrong follows.*
 - A lexical entry for dummy *it*
 - A rule to account for the relationship between pairs like (8) and (9)
 - (9) *It follows that you are wrong.*

*We need these anyway (independently motivated)

The Entry for Dummy *it*

$\langle$ it,	<i>pron-lxm</i>		
	SYN	HEAD	$\left[\begin{array}{ll} \text{FORM} & \textit{it} \\ \text{AGR} & \textit{3sing} \end{array} \right]$
	SEM	MODE	<i>none</i>
		INDEX	<i>none</i>
		RESTR	$\langle \rangle$

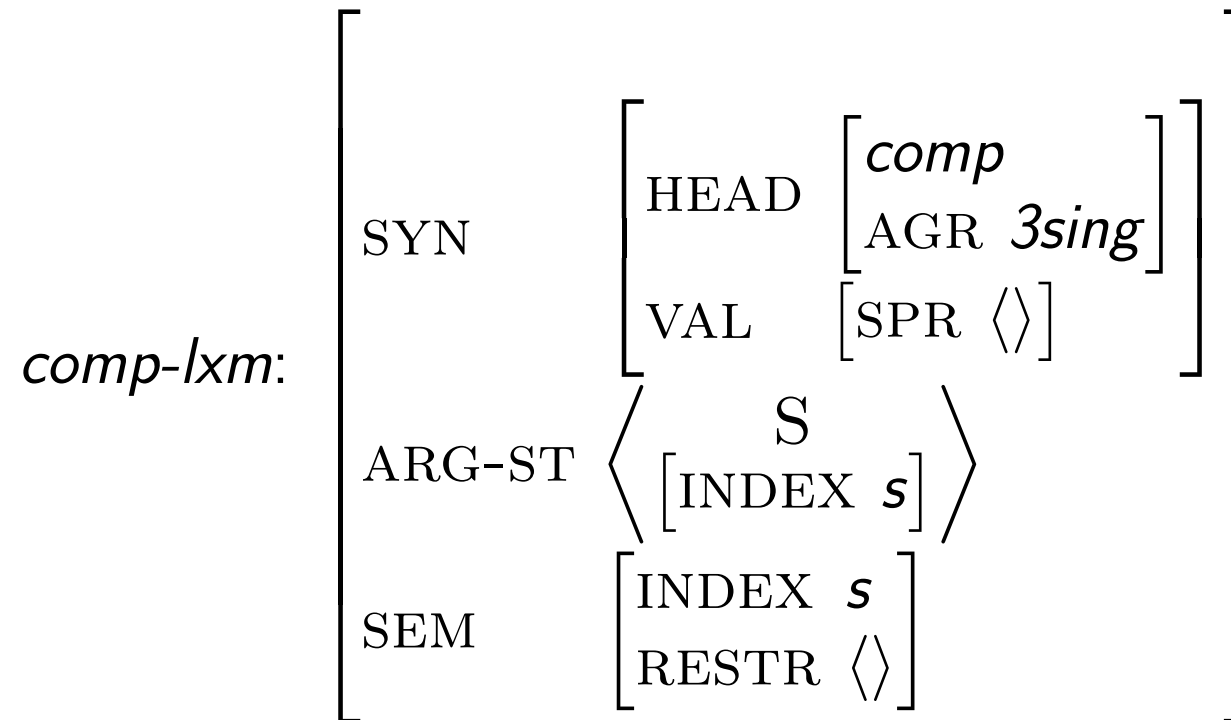
Questions About Dummy *it*

- How does it differ from the entry for dummy *there*? Why do they differ in this way?
- Is this the only entry for *it*?

$\langle$ it,	<i>pron-lxm</i>		
	SYN	HEAD	$\left[\begin{array}{ll} \text{FORM} & \textit{it} \\ \text{AGR} & \textit{3sing} \end{array} \right]$
	SEM	MODE	<i>none</i>
		INDEX	<i>none</i>
		RESTR	$\langle \rangle$

Complementizers and CP Subjects

A New Type of Lexeme: Complementizers

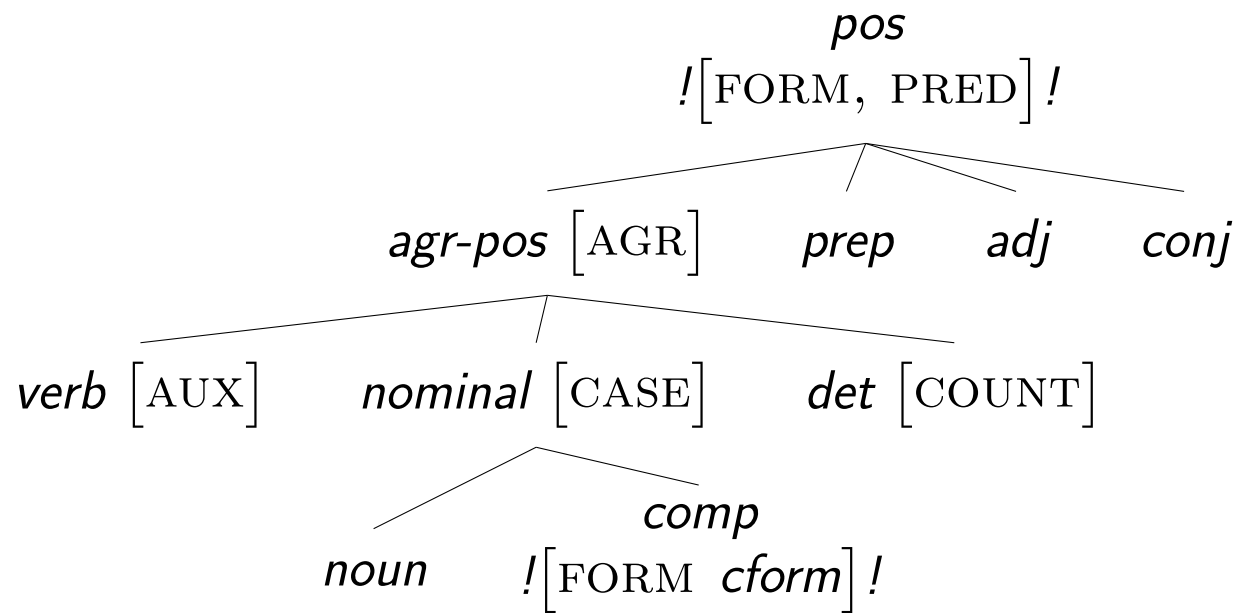


Questions About the Type *comp-lxm*

- Why does it stipulate values for both SPR and ARG-ST?
- Why is its INDEX value the same as its argument's?
- What is its semantic contribution?

$$\text{comp-lxm:} \left[\begin{array}{l} \text{SYN} \left[\begin{array}{l} \text{HEAD} \left[\begin{array}{l} \text{comp} \\ \text{AGR } 3\text{sing} \end{array} \right] \\ \text{VAL} \left[\begin{array}{l} \text{SPR } \langle \rangle \end{array} \right] \end{array} \right] \\ \text{ARG-ST} \left\langle \begin{array}{c} \text{S} \\ \left[\text{INDEX } s \right] \end{array} \right\rangle \\ \text{SEM} \left[\begin{array}{l} \text{INDEX } s \\ \text{RESTR } \langle \rangle \end{array} \right] \end{array} \right]$$

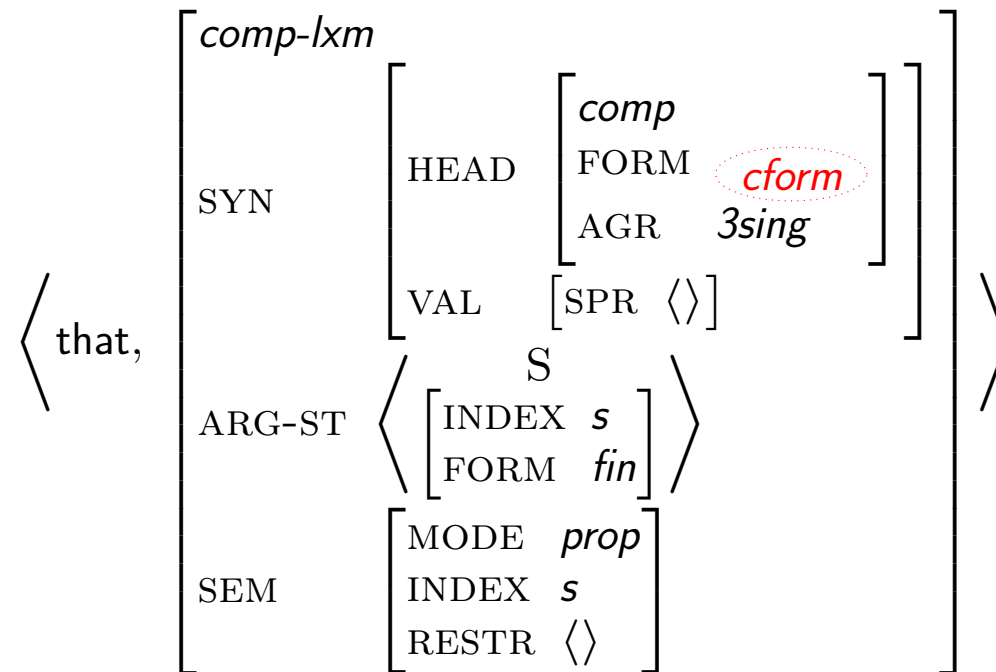
The Type *comp*



The Lexical Entry for Complementizer *that*

$$\left\langle \text{that}, \begin{bmatrix} \textit{comp-lxm} \\ \text{ARG-ST} \left\langle \begin{bmatrix} \text{FORM} \textit{fin} \end{bmatrix} \right\rangle \\ \text{SEM} \quad \begin{bmatrix} \text{MODE} \textit{prop} \end{bmatrix} \end{bmatrix} \right\rangle$$

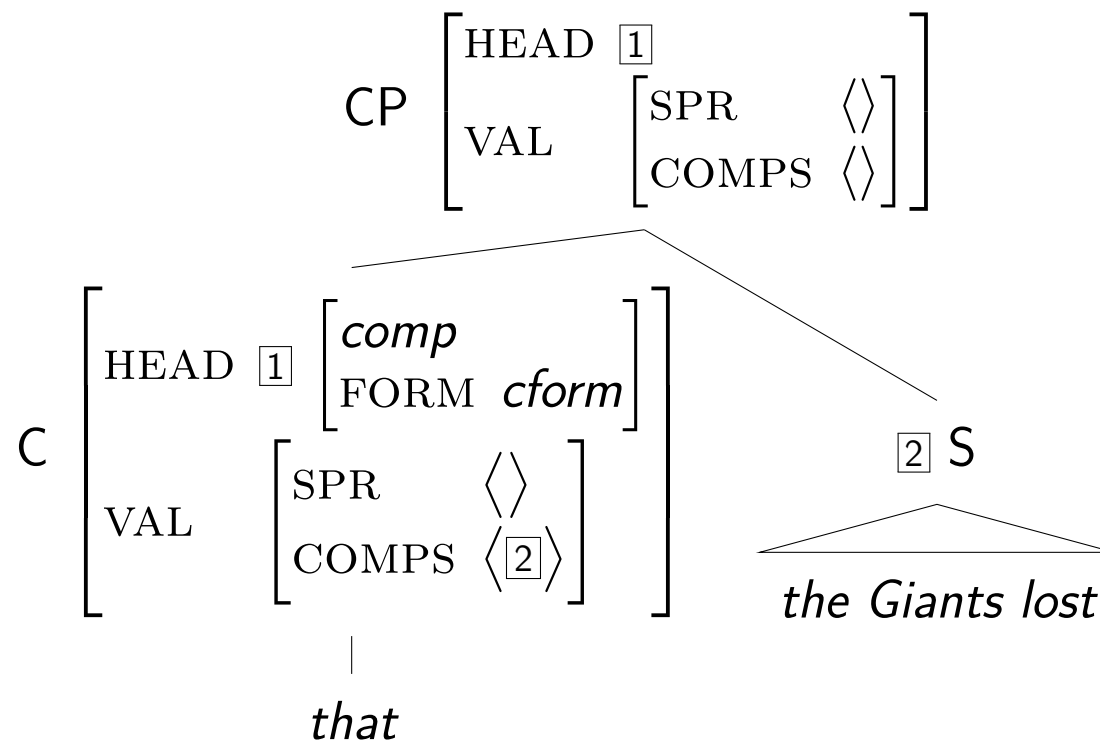
...with Inherited Information Filled In



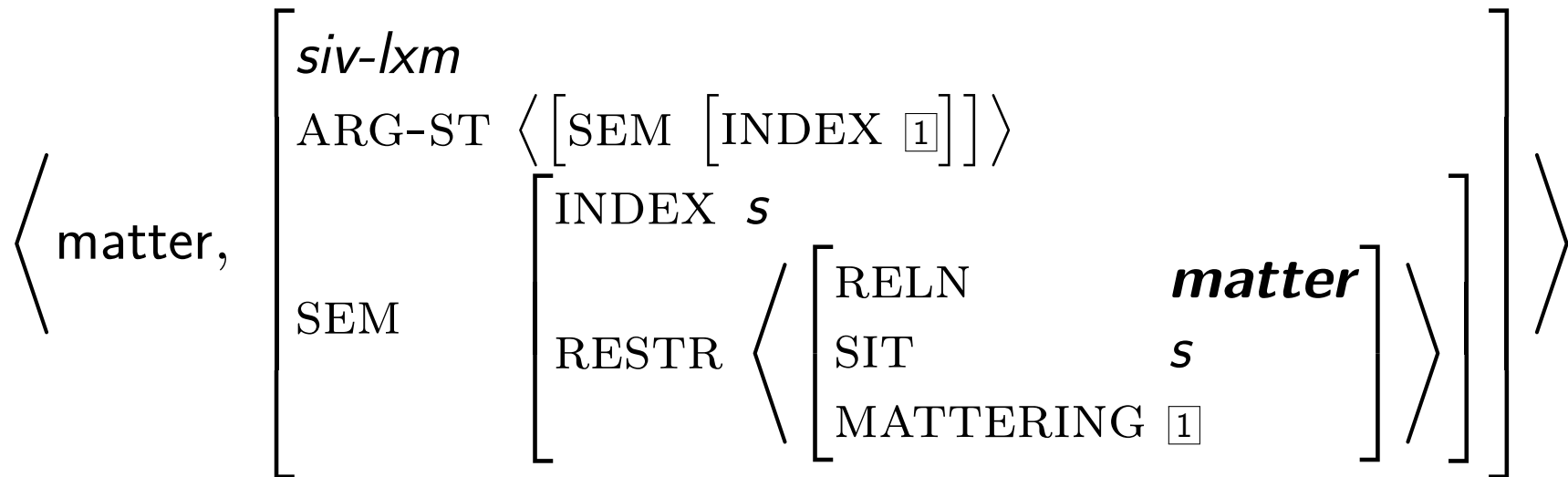
- Question: Where did [FORM *cform*] come from?

Structure of a Complementizer Phrase

(10) *that the Giants lost*



Sample Verb with a CP Subject



Note: the only constraint on the first argument is semantic.

A Problem

- We constrained the subject of *matter* only semantically. However...

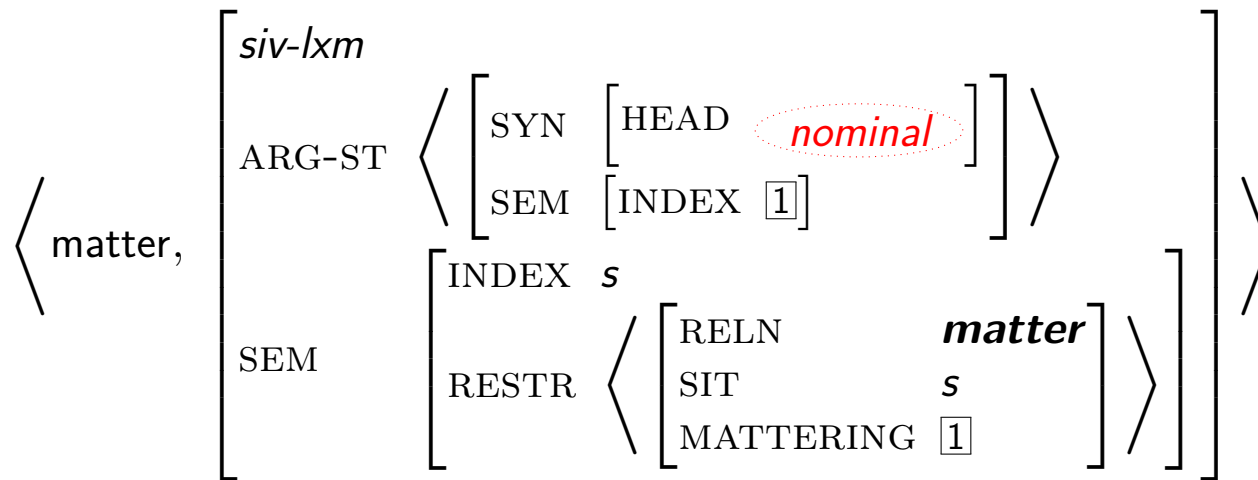
- CP and S are semantically identical, but we get:

(11) *That Bush won matters* vs. **Bush won matters*

- Argument-marking PPs are semantically identical to their object NPs, but we get:

(12) *The election mattered* vs. **Of the election mattered*

- So we need to add a syntactic constraint.



- S and PP subjects are generally impossible, so this constraint should probably be on *verb-lxm*.

Extraposition

Extraposition (at Last)

- Extraposition alters word order so that a relatively “heavy” constituent appears to the right of its canonical position.

- (13) a. That you were wrong follows.
b. It follows that you were wrong.
- (14) a. That I mistyped it was frustrating.
b. It was frustrating that I mistyped it.
- (15) a. Did that this happened surprise you?
b. Did it surprise you that this happened?

Post-Inflectional Lexical Rule: *pi-rule*

$$\begin{array}{c}
 \textit{pi-rule:} \left[\begin{array}{c} \textit{l-rule} \\ \text{INPUT} \left\langle / \boxed{0}, \left[\begin{array}{c} \textit{word} \\ \text{SYN} \left[\begin{array}{c} \text{HEAD} / \boxed{1} \\ \text{VAL} \left[\text{MOD } \boxed{A} \end{array} \right] \end{array} \right] \right\rangle \\ \text{OUTPUT} \left\langle / \boxed{0}, \left[\begin{array}{c} \textit{word} \\ \text{SYN} \left[\begin{array}{c} \text{HEAD} / \boxed{1} \\ \text{VAL} \left[\text{MOD } \boxed{A} \end{array} \right] \end{array} \right] \right\rangle \end{array} \right]
 \end{array}$$

We use this when we expect the structure to be largely formed, and we want to change the order of things, ...

The Extraposition Lexical Rule

$$\left[\begin{array}{l} \textit{pi-rule} \\ \text{INPUT} \left\langle X, \left[\text{SYN} \left[\text{VAL} \left[\begin{array}{l} \text{SPR} \quad \langle \boxed{1} \text{ CP} \rangle \\ \text{COMPS} \quad \boxed{A} \end{array} \right] \right] \right] \right\rangle \\ \text{OUTPUT} \left\langle Y, \left[\text{SYN} \left[\text{VAL} \left[\begin{array}{l} \text{SPR} \quad \langle \text{NP} [\text{FORM } it] \rangle \\ \text{COMPS} \quad \boxed{A} \oplus \langle \boxed{1} \rangle \end{array} \right] \right] \right] \right\rangle \end{array} \right]$$

- Why is the type *pi-rule*?
- Why doesn't it say anything about the semantics?
- Why is the COMPS value $\boxed{A}$ not $\langle \rangle$?

Extraposition with Verbs Whose comps Lists Are Nonempty

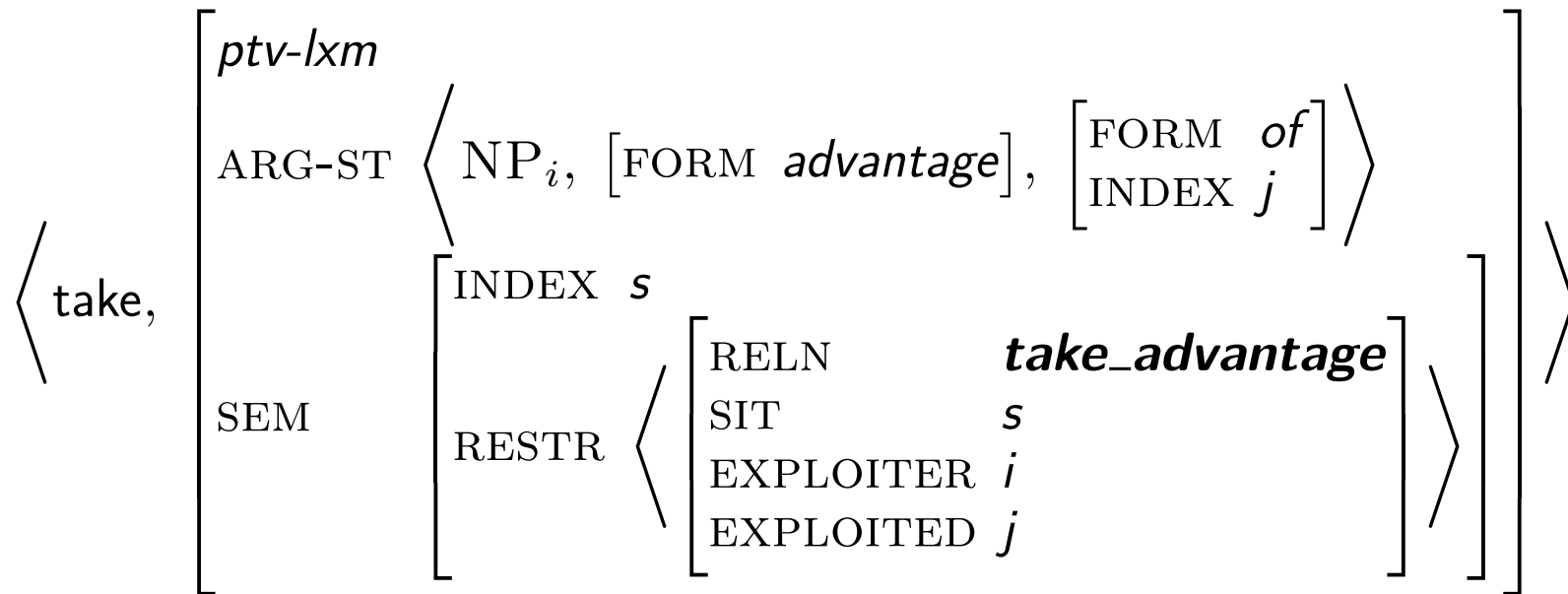
- (16) *It worries me that war is imminent.*
- (17) *It occurred to Pat that Chris knew the answer.*
- (18) *It endeared you to Andy that you wore a funny hat.*

Idioms

Another Nonreferential Noun: *advantage*

$\langle$ advantage,	$\left[\begin{array}{l} \textit{massn-lxm} \\ \text{SYN} \left[\begin{array}{l} \text{HEAD} \left[\begin{array}{l} \text{FORM} \textit{advantage} \\ \text{AGR} \textit{3sing} \end{array} \right] \\ \text{SEM} \left[\begin{array}{l} \text{MODE} \textit{none} \\ \text{INDEX} \textit{none} \\ \text{RESTR} \langle \rangle \end{array} \right] \end{array} \right] \end{array} \right]$			$\rangle$

The Verb That Selects *advantage*



Our Analyses of Idioms and Passives Interact...

- We generate

(19) *Advantage was taken of the situation by many people.*

(20) *Tabs are kept on foreign students.*

with roughly the same meanings as

(21) *Many people took advantage of the situation.*

(22) *Someone keeps tabs on foreign students.*

- But not:

(23) *Many people were taken advantage of.*

- Why not?

Overview

- Review: the copula *be*
- Existentials (*there*, *be*)
- Dummy *it*
- Complementizers and CP subjects (*that*)
- Extraposition (*it*, LR)
- Idioms (*take_advantage*, ...)
- Problems

Problems

P1: *there* and Agreement

The analysis of existential *there* sentences presented so far says nothing about verb agreement.

- A. Consult your intuitions (and/or those of your friends, if you wish) to determine what the facts are regarding number agreement of the verb in *there* sentences.

Consider these sentences: which ones do you think are ok?

- (24) There is a cat on the mat.
- (25) There are a cat on the mat.
- (26) There is cats on the mat.
- (27) There are cats on the mat.

Give an informal statement of a generalization covering these facts, and illustrate it with both grammatical and ungrammatical examples. [*Note: Intuitions vary regarding this question, across both individuals and dialects. Hence there is more than one right answer to this question.*]

For me (Francis), only (24) and (27) are ok; for many speakers (26) is also ok.

- B. How would you elaborate or modify our analysis of the *there* construction so as to capture the generalization you have discovered? Be as precise as you can.

P2: Passing Up the Index

A. Give the RESTR value that our grammar should assign to the sentence in (i). Be sure that the SIT value of the **smoke** predication is identified with the ANNOYANCE value of the **annoy** predication.

(i) *That Dana is smoking annoys Leslie.*

[Hint: This sentence involves two of the phenomena analyzed in this chapter: predicative complements of **be** and CP subjects.]

B. Draw a tree for (i). Use abbreviations for node labels, but show the index on each node.

C. Explain how the SIT value of the **smoke** predication gets identified with the ANNOYANCE value of the **annoy** predication. Be sure to make reference to lexical entries, phrase structure rules, and principles, as appropriate.

P3: An Annoying Problem

Assume that the lexical entry for the verb *annoy* is the following:

$$(28) \left\langle \text{annoy} , \left[\begin{array}{l} \textit{stv-lxm} \\ \text{ARG-ST} \left\langle \left[\text{SEM} \left[\text{INDEX} \boxed{1} \right] \right], \text{NP}_i \right\rangle \\ \text{SEM} \left[\begin{array}{l} \text{INDEX } s \\ \text{RESTR} \left\langle \left[\begin{array}{l} \text{RELN} \quad \textit{annoy} \\ \text{SIT} \quad s \\ \text{ANNOYED} \quad i \\ \text{ANNOYANCE} \quad \boxed{1} \end{array} \right] \right\rangle \end{array} \right] \end{array} \right] \right\rangle$$

-
- A. What constraints are imposed on the lexical sequences that result from applying the **3rd-Singular Verb Lexical Rule** to this entry (including those that involve inheritance of constraints from the entry's supertypes)?
- B. What constraints are imposed on lexical sequences that result from applying the **Extraposition Lexical Rule** to your answer to part (A)?
- C. Draw a tree structure for the sentence in (29). You should show the value of all SEM features on all of the nodes, as well as the SPR and COMPS features for *annoys*.

(29) *It annoys Lee that Fido barks.*

D. The lexical entry for *annoy* allows NP subjects as well, as in (30). Why doesn't the grammar then also license (31)?

(30) *Sandy annoys me.*

(31) **It annoys me Sandy.*

Acknowledgments and References

- Course design and slides borrow heavily from Emily Bender's course: *Linguistics 566: Introduction to Syntax for Computational Linguistics*
<http://courses.washington.edu/ling566>
- This lecture complements Sag et al. (2003, Chapter 11)



References

Ivan A. Sag, Tom Wasow, and Emily Bender. 2003. *Syntactic Theory: A Formal Introduction*. CSLI Publications, Stanford, second edition.